Markforged

AEROSPACE · CERTIFIED ADDITIVE MANUFACTURING



Not just a part — a pathway.

Plane Support built a dedicated aerospace additive manufacturing laboratory around the Markforged FX10 — and gained CASA approval to manufacture cabin air conditioning ducting for McDermott Aviation's Bell 214B.

PLANE SUPPORT × MCDERMOTT AVIATION · QUEENSLAND, AUSTRALIA · 2026

APMA
CASA parts manufacturer approval
214B (B-1) cabin A/C ducting

~2 yrs
of development and approval work with CASA

29
Bell 214s operated by McDermott
15 × 214B + 14 × 214ST

FX10
core platform of the dedicated aerospace AM lab

01

The Challenge

PARTS YOU CAN'T GET HOLD OF

APPROVAL HOLDER Plane Support — Australian aviation engineering, manufacturing and maintenance; serving the industry for almost 30 years	INITIATED BY McDermott Aviation — Australia's largest privately owned helicopter operator, Sunshine Coast, Queensland
AIRCRAFT Bell 214B (B-1) · McDermott holds the Type Certificate for both the 214B and 214ST	PART Cabin air conditioning ducting
FRAMEWORK CASA Australian Parts Manufacturer Approval (APMA) · expands existing CASA Part 21K AM capability	PLATFORM Markforged FX10 · dedicated aerospace additive manufacturing laboratory

The Bell 214B is no longer in production. McDermott Aviation operates the world's largest collection of the type — 15 × 214B and 14 × 214ST — flying aerial firefighting, heavy lifting, agricultural spraying, infrastructure support and emergency response. At that fleet scale, **a component you cannot source is not an inconvenience; it is an aircraft on the ground**.

And the responsibility has nowhere else to go. McDermott holds the **Type Certificate** for both the 214B and the 214ST — long-term support for the type sits with the operator itself.

> **“An aircraft waiting on a component you can't easily get hold of is an aircraft that isn't working. Being able to have approved parts made here in Queensland gives us far more control over that.”**
>
> — **John McDermott**, Founder & President, McDermott Aviation



The class of part approved: Bell 214B cabin ducting, printed on the Markforged FX10. Photo: Plane Support

02

The Solution

SCAN, REVERSE ENGINEER, PROVE IT

Plane Support took the engineering, manufacturing and approvals side; McDermott brought decades of operating knowledge; Markforged's Australian reseller partner Invenio provided local support. It took **almost two years**, in direct consultation with CASA, to take one duct from original component to approved part.

1 → 2 → 3 → 4

3D Scanning

Capture the original 214B ducting, build accurate digital models

Reverse Engineering

Redesign for AM; structure validated in Markforged Simulation

Dedicated Lab

Aerospace AM laboratory built around the Markforged FX10

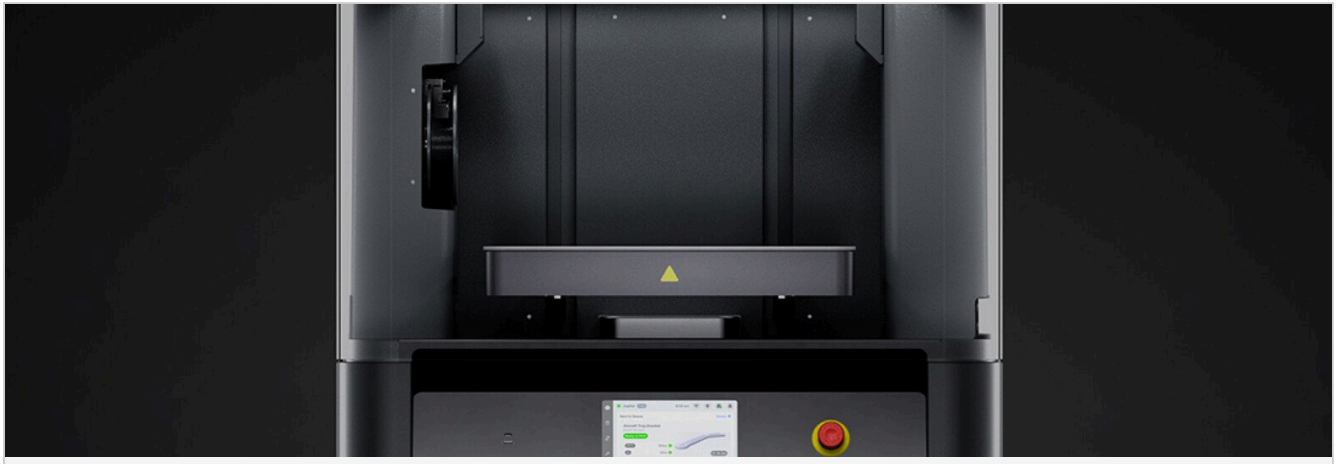
Proving It

Quality systems, production controls and inspection methods, evidenced to CASA

What are APMA and Part 21K?

APMA (Australian Parts Manufacturer Approval) is CASA's approval for a parts manufacturer: it allows an organisation other than the original manufacturer to legally produce specified aircraft parts for installation. The approval binds who makes it, what they make, and how.

Part 21K is the airworthiness category covering parts manufacture. This approval expands Plane Support's existing Part 21K additive manufacturing capability — meaning the capability was never built for a single part.



The platform: Plane Support established a dedicated aerospace additive manufacturing laboratory for the project, built around the **Markforged FX10** industrial platform. Image: Markforged official.

> **“Ducting sounds simple until you try to reproduce it. These are complex shapes that have to fit exactly and perform in the aircraft just as the originals do, and we had to prove every step of that to CASA.”**
>
> — **Jason Hazell**, Director of Engineering, Plane Support

03

What Was Approved

SCOPE, PRECISELY

Scope of the CASA Approval

WHO, WHAT AND UNDER WHICH FRAMEWORK

Holder	**Plane Support** — as the parts manufacturer
Part	Bell **214B (B-1) cabin air conditioning ducting**
Framework	CASA **Australian Parts Manufacturer Approval (APMA)**
Effect	Expands Plane Support's existing **CASA Part 21K** additive manufacturing capability
Standing	Part 21K · Part 21G · Part 145 approvals, operating an **AS9100** quality management system

APMA
Approved to manufacture and install

21K
Existing Part 21K scope expanded by this project

~2 yrs
Development and evidence, in consultation with CASA

1 → n
The same groundwork extends to other parts and types

What this approval really produced is not the duct. It is **a route that has been walked once** — and the next part takes the same route.

> **“We didn't have an established process to follow, so we developed our own procedures, quality systems and inspection methods and worked them through with CASA. That groundwork now gives us a pathway for other components and other aircraft types.”**
>
> — **Jason Hazell**, Director of Engineering, Plane Support

A note on wording

CASA's approval applies to Plane Support as a manufacturer and to the manufacturing solution for that specific part; the Markforged FX10 is the platform used within the approved process. The precise formulation is “**a process built around the Markforged FX10, approved by CASA under APMA**”. Neither release discloses lead times or costs; none are estimated here.

04

Significance & Lineage

WHY IT TRAVELS FURTHER THAN ONE DUCT

For **fleet operators and Type Certificate holders**, out-of-production parts now have a lawful local supply route that does not depend entirely on the OEM supply chain's pace. For **MROs and parts manufacturers**, APMA and Part 21K are qualifications that accumulate — the first part is the hard one; after that it is reuse. The case also makes a wider point: **the barrier to airworthiness evidence sits in procedure and inspection, not in the printer**.

Markforged platforms have an existing track record in certified civil aviation parts:

2024	**Spectrum Networks × ALOFT AeroArchitects (USA)** Over 10,000 certified parts produced in a single year and flying on aircraft today, on FX10 and X7 platforms; laser micrometer inspection layer by layer with automated conformance reporting. VIP and head-of-state cabins, and obsolete part replacement.
Qualified	**Cabin Management Solutions (USA)** Onyx FR-A and Carbon Fiber FR-A qualified on the X7 through the NCAMP process; 100+ end-use flight parts developed with full material traceability.
Materials	**Flame-retardant composites and continuous fibre** Onyx FR-A / CF FR-A address FAA 14 CFR 25.853 cabin flammability requirements; continuous fibre reinforcement delivers aluminium-class strength at roughly one third of the weight.
2026	**Plane Support × McDermott Aviation (Australia · this case)** A process built around the FX10 gains CASA APMA approval for Bell 214B cabin air conditioning ducting; Onyx FR-A material, validated in Markforged Simulation before printing.



The weight of the Type Certificate: John McDermott, Founder & President — “We own the Type Certificate for this aircraft, so keeping it supported sits with us.”

05

What Makes the Evidence Hold

MATERIAL · INSPECTION · SIMULATION

Plane Support's own procedures, quality systems and inspection methods are the substance of the submission. Markforged's official case study (28 October 2026) names three elements that hold it up: **Onyx FR-A** lot-level traceability, Plane Support's **non-destructive inspection methods** under Part 21K, and **Markforged Simulation** before printing.

Material	**Onyx FR-A — lot-level material traceability** Markforged's aerospace-grade, flame-retardant, micro-carbon-fibre-filled nylon, with **complete lot-level material traceability** — the documentation and repeatability airworthiness frameworks require. FR-A addresses **FAA 14 CFR 25.853** and is NCAMP-qualified on the X7; **the FX10 runs the same material**.
Inspection	**Non-destructive inspection — built by Plane Support under Part 21K** With no precedent to follow, Plane Support built the procedures, quality controls and **non-destructive inspection methods** required under **CASA Part 21K** from the ground up — its own qualification asset, and its route to other parts and airframes.
Simulation	**Markforged Simulation — validating structure before printing** The complex internal duct geometry required strict structural verification; the team used **Markforged Simulation** to analyse and optimise part performance virtually before printing.

> **“This milestone from Plane Support and McDermott Aviation marks a defining moment for additive manufacturing across the global aerospace sector. By pairing the industrial precision of the FX10 with the predictive power of Markforged Simulation and the lot-level traceability of our aerospace-grade Onyx FR-A, they have addressed a critical supply bottleneck head-on. They have proven that flight-ready, fully certified composite components can be produced predictably and locally to keep mission-critical aircraft in the air.”**
>
> — **Richard Elving**, Head of Sales - APAC, Markforged

A note on plan tiers

Markforged Simulation is a module of Digital Forge Complete (DFC) Advanced — it is not bundled with the printer, and the On-Prem (air-gapped) configuration differs in scope. Confirm plan tier and printer support before deployment.

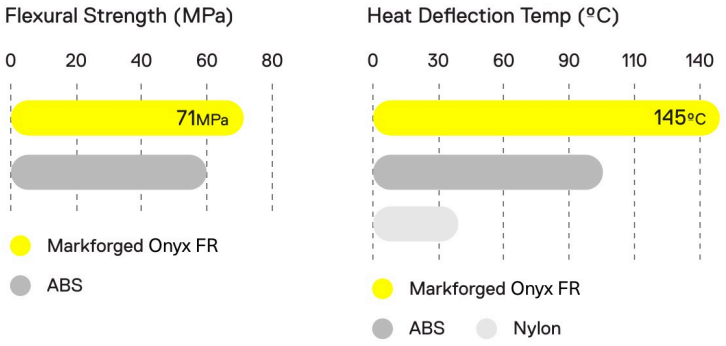
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Material Profile: Onyx FR-A

AEROSPACE-READY COMPOSITE

The ducting is printed in **Onyx FR-A**, the aerospace variant of Markforged's flame-retardant composite Onyx FR. It is the same micro-carbon-fibre-filled, flame-retardant nylon, with the two things an aircraft part needs added: **lot-level material traceability**, and a pass on the test suite required under **14 CFR 25.853** for most 3D-printable parts. Markforged positions it as purpose-built for the aerospace, transportation and automotive industries.





Material comparison: flexural strength and heat deflection temperature against ABS and nylon. The chart is labelled Onyx FR; the official datasheet gives the same two values for Onyx FR-A. Chart: Markforged customer case study

71 MPa Flexural strength ASTM D790	**145** °C Heat deflection temp ASTM D648 B	**3.6** GPa Flexural modulus ASTM D790	**40** MPa Tensile strength ASTM D638

FST

Flame, smoke and toxicity

At 2 mm, passes the FAR 25.853 vertical (60 s and 12 s) and horizontal burn tests and Boeing BSS 7239 combustion toxicity; smoke density passes at 3.7 mm. Heat release did not pass on the 3.7 mm specimen — under FAA policy statement PS-ANM-25.853-01-R2, most interior cabin parts with exposed surface area below the threshold do not require that test. Whether a given part qualifies still depends on the part.

Printers

X7, FX10 and FX20, with continuous carbon fibre

All three print Onyx FR-A and can reinforce it with **Carbon Fiber FR-A** continuous carbon fibre; the datasheet lists a flexural strength of 540 MPa for Carbon Fiber FR-A. The official case study does not say whether the ducting uses continuous fibre.

Values from the Markforged “Onyx FR-A and Carbon Fiber FR-A Supplemental Datasheet”, REV 1.2A (16 June 2025). Results vary with environmental conditions, feature geometry, print orientation and loading conditions.

Markforged

Printing it was never the hard part. Proving it can fly was.

Certify once. Reuse the pathway.

Plane Support spent almost two years taking a single Bell 214B duct from 3D scan, through reverse engineering and redesign, all the way to a regulator's signature. With the procedures, quality systems and inspection methods in place, the next part — and the next aircraft type — takes the same route. That is why this approval is worth considerably more than the part it covers.

SOURCES

Markforged — “Markforged Technology Powers Milestone CASA Approval for Flight-Ready 3D-Printed Components on Heavy-Lift Firefighting Helicopters” (28 October 2026)

Markforged Customer Success Story — Plane Support (October 2026)

Markforged — Onyx FR-A and Carbon Fiber FR-A Supplemental Datasheet, REV 1.2A (16 June 2025)

AerialFire Magazine — “Plane Support Gains CASA Approval for 3D Printed 214B Helicopter Parts” (12 October 2026)

Plane Support — planesupport.com · McDermott Aviation — mcdermottaviation.com · Invenio — invenio3d.com.au

Note: facts, titles, quotations, regulatory framework and timelines are taken from Markforged's official release and customer case study of 28 October 2026, and from the release issued jointly by Plane Support and McDermott Aviation on 12 October 2026 (fleet numbers come from the latter). CASA's approval applies to Plane Support as a manufacturer and to its specific part manufacturing solution; it is not an airworthiness endorsement of any printer brand. Neither release discloses lead times or costs, and none are estimated here. Material values are from the Onyx FR-A supplemental datasheet, REV 1.2A. People and part photography: Plane Support.